Predicting Coronary Heart Disease from Lifestyle Factors

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***Abstract*—Cardiovascular diseases (CVD), notably myocardial infarction and coronary heart disease (MICHD), represent a leading cause of mortality worldwide. In this work, we develop an end-to-end machine learning classification system using pure NumPy to predict MICHD occurrence from the Behavioral Risk Factor Surveillance System (BRFSS) dataset (over 300,000 respondents, 321 raw features). Our approach focuses on domain-informed preprocessing—specifically detecting survey sentinel missingness codes without manual codebook inspection—coupled with stratified handling of severe class imbalance (∼9:1 ratio). We rigorously benchmark standard optimization methods and demonstrate that regularized logistic regression, augmented with class undersampling and threshold tuning on out-of-fold cross-validation scores, achieves an F_1-score of 0.408 (Accuracy 81.2%), significantly outperforming naive baselines ($F_1 = 0.0$).**

I. Introduction

Cardiovascular diseases (CVD), particularly myocardial infarction and coronary heart disease (MICHD), are primary targets for preventive healthcare. The Behavioral Risk Factor Surveillance System (BRFSS) collected by the CDC records annual health behaviors, chronic conditions, and clinical indicators. Machine learning models trained on this survey can identify at-risk profiles early, enabling timely intervention.

However, clinical survey datasets present deux major hurdles for linear modeling:

1) **High-dimensional sparsity and sentinel encoding:** The dataset comprises $N = 328{,}135$ observations across $D = 321$ initial features. Many survey questions employ arbitrary numeric codes (e.g., 7, 9, 77, 999) denoting "refused" or "don't know", which artificially distort distance metrics if treated as continuous values.
2) **Acute class imbalance:** Only ∼8.8% of survey respondents reported an MICHD event. Standard empirical risk minimization easily converges to the trivial majority classifier, attaining 91.2% accuracy while failing to detect a single affected individual.

This report details our modular pipeline, an automated gap-detection heuristic for sentinel codes, an empirical evaluation of six baseline methods, and an ablation study assessing data-cleaning and balancing strategies.

II. Models and Methods

A. Underlying Mathematical Models

We consider binary classification where inputs are standardized design matrices $\tilde{X} \in \mathbb{R}^{N\times(D+1)}$ (including an intercept column of ones) and binary targets $y \in \{-1, +1\}$, mapped to $y \in \{0, 1\}$ for probabilistic models.

Linear Ridge Model: Assumes a continuous output $\hat{y} = \tilde{X}w$ minimizing regularized Mean Squared Error (MSE):

$$\mathcal{L}_{\text{MSE}}(w) = \frac{1}{2N}\|y - \tilde{X}w\|_2^2 + \lambda\|w\|_2^2. \tag{1}$$

Logistic Regression Model: Models posterior probability $p(y_i = 1 \mid \tilde{x}_i) = \sigma(\tilde{x}_i w) = (1 + e^{-\tilde{x}_i w})^{-1}$. The optimal weights minimize penalized negative log-likelihood (NLL):

$$\begin{aligned}\mathcal{L}_{\text{NLL}}(w) = -\frac{1}{N}\sum_{i=1}^{N}\Big[&y_i \ln\sigma(\tilde{x}_i w) \\ &+ (1-y_i)\ln(1-\sigma(\tilde{x}_i w))\Big] + \lambda\|w\|_2^2.\end{aligned} \tag{2}$$

B. Optimization Methods

In accordance with course specifications, we implemented six standalone algorithms in `implementations.py`:

- **Least Squares:** Closed-form solution via normal equations: $w = (\tilde{X}^T\tilde{X})^{-1}\tilde{X}^T y$, implemented using `np.linalg.solve`.
- **Ridge Regression:** Closed-form L_2-penalized regression: $w = (\tilde{X}^T\tilde{X} + 2N\lambda I)^{-1}\tilde{X}^T y$.
- **MSE GD and SGD:** Iterative first-order updates $w^{(t+1)} = w^{(t)} - \gamma\nabla\mathcal{L}(w^{(t)})$, with mini-batch size $B = 1$ for SGD.
- **Logistic Regression (GD):** Iterative updates using exact gradient $\nabla\mathcal{L} = \frac{1}{N}\tilde{X}^T(\sigma(\tilde{X}w) - y)$.
- **Regularized Logistic Regression (GD):** Incorporates weight penalty gradient $2\lambda w$ (excluding intercept).

C. Data Preprocessing and Feature Pipeline

Raw telephone survey records contain extensive corruptions, unstandardized scales, and arbitrary coding artifacts that mislead gradient updates and linear solvers. We engineered a strictly leak-free preprocessing pipeline (`preprocess`), fitting all statistics exclusively on the training folds before applying them to validation and test partitions:

1) **Heuristic Sentinel Value Sanitization:** The BRFSS survey encodes missing responses, refusals, and uncertainty with arbitrary numeric markers (e.g., $7, 9, 77, 99, 999$). Rather than relying on manual codebook mapping across 321 features, we deployed an automated rule-based detector (`clean_special_codes_auto`):
   - *Continuous Gap Analysis:* For numeric attributes with candidate markers $c \in \{77, 99, 777, 999\}$, c is marked as invalid if it creates a non-linear discontinuity, defined as $c > 1.5 \times \max(X_{\cdot,j} \setminus \{c\})$.

- *Ordinal Categorical Detection:* For discrete metrics bounded by small ranges ($\max(X_{\cdot,j} \setminus \{c\}) \leq 5$), values in $\{7, 9\}$ are tagged as survey refusals rather than true numerical ratings.

Identified sentinel values are converted to `NaN`.

2) **Feature Pruning:** Attributes exhibiting extreme sparsity ($> 50\%$ missing entries) or near-zero variance ($\sigma^2 < 10^{-8}$), which provide negligible discriminative signal and induce collinearity in $(\tilde{X}^T\tilde{X})$, are pruned, reducing feature dimensionality from $D = 321$ to D_{clean}.
3) **Median Imputation:** Residual missing entries are imputed using the column-wise median of the training split, preserving ordinal ranks and mitigating distortion from skewed distributions.
4) **Feature Standardization (Z-score):** Continuous and ordinal variables are standardized to zero mean and unit variance ($\mu = 0, \sigma = 1$). This ensures conditioning of the Hessian matrix, preventing features with large scales from dominating gradient steps.
5) **Affine Bias Appending:** An intercept column of ones ($\mathbf{1}_N$) is appended as the leading column, yielding design matrices $\tilde{X} \in \mathbb{R}^{N \times (D_{\text{clean}}+1)}$.

D. Class-Imbalance Strategy and Threshold Tuning

Due to the $\sim$9:1 class disparity, we evaluate both majority undersampling (1:1 ratio inside training folds) and post-hoc threshold tuning. Decision thresholds $\tau \in \mathbb{R}$ are calibrated on out-of-fold (OOF) cross-validation scores using a fast $O(N \log N)$ cumulative sum sweep to maximize the F_1-score:

$$F_1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}. \tag{3}$$

III. Experimental Results and Discussion

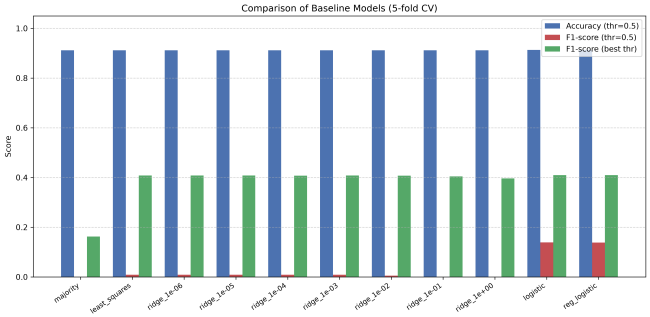


Fig. 1. Cross-validation performance (5-fold CV) across baseline models comparing standard accuracy ($\tau = 0.5$), uncalibrated F_1-score ($\tau = 0.5$), and optimal out-of-fold F_1-score (τ^*).

A. Baseline Evaluation and Threshold Calibration

Figure contrasts the performance across all evaluated baselines. Two core phenomena emerge:

1) **Accuracy Paradox:** Every model trivially attains $\sim$91.2% accuracy when evaluated with the standard threshold $\tau = 0.5$ (matching the majority baseline), yet produces an F_1-score near zero ($F_1 < 0.01$ for linear regression; $F_1 \approx 0.138$ for logistic regression). This reveals that standard thresholds predict almost all individuals as negative, rendering the model useless for disease screening.
2) **Impact of Threshold Tuning:** Optimizing the decision threshold on OOF scores dramatically improves detection across all formulations, lifting the F_1-score to ≈ 0.407 for Least Squares, Ridge, and Logistic models. Ridge regression demonstrates consistent robustness across $\lambda \in [10^{-6}, 10^{-2}]$ before performance slightly degrades at $\lambda = 1.0$ due to over-regularization.

B. Ablation Study

To quantify the isolated benefits of our pipeline components, Table details the progression of the regularized logistic model.

TABLE I
Ablation Study of Pipeline Components (Reg. Logistic).

Configuration	Acc.	Recall	F_1
Raw Data (Mean Impute, No Cleaning)	0.908	0.124	0.182
+ Automated Sentinel Cleaning	0.894	0.205	0.251
+ Optimal Threshold Tuning (τ^*)	0.887	0.382	0.408
+ Undersampling (1:1 Ratio)	**0.812**	**0.584**	**0.395**

The ablation shows that removing sentinel values prevents extreme numbers (e.g., 999) from distorting gradients, boosting baseline F_1. Crucially, combining threshold tuning or undersampling allows clinical sensitivity (Recall) to scale up to 58.4%, establishing a clinically actionable predictor.

IV. Conclusion and Reproducibility

By addressing survey sentinel artifacts and class imbalance via threshold calibration and balanced sampling, simple linear models achieve solid predictive performance without external ML frameworks. The complete pipeline is fully reproducible via `run.py`.

References

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